

Experiment 5: generalization of the Caesar cipher

```
#include <stdio.h>

#include <string.h>

#include <ctype.h>


int main()
{
    char text[100];

    int a, b, i, p, c;


    printf("Enter plaintext: ");
    scanf("%[^\n]", text);


    printf("Enter value of a: ");
    scanf("%d", &a);


    printf("Enter value of b: ");
    scanf("%d", &b);


    if (a < 0 || a > 25 || b < 0 || b > 25)
    {
        printf("Invalid values of a or b\n");
        return 0;
    }


    int gcd = 1;
    for (i = 1; i <= a && i <= 26; i++)
    {
        if (a % i == 0 && 26 % i == 0)
            gcd = i;
```

```
}
```

```
if (gcd != 1)
```

```
{
```

```
    printf("Invalid value of a. GCD(a, 26) must be 1.\n");
```

```
    return 0;
```

```
}
```

```
printf("Encrypted text: ");
```

```
for (i = 0; text[i] != '\0'; i++)
```

```
{
```

```
    if (isalpha(text[i]))
```

```
    {
```

```
        if (isupper(text[i]))
```

```
            p = text[i] - 'A';
```

```
        else
```

```
            p = text[i] - 'a';
```

```
        c = (a * p + b) % 26;
```

```
        if (isupper(text[i]))
```

```
            printf("%c", c + 'A');
```

```
        else
```

```
            printf("%c", c + 'a');
```

```
    }
```

```
    else
```

```
    {
```

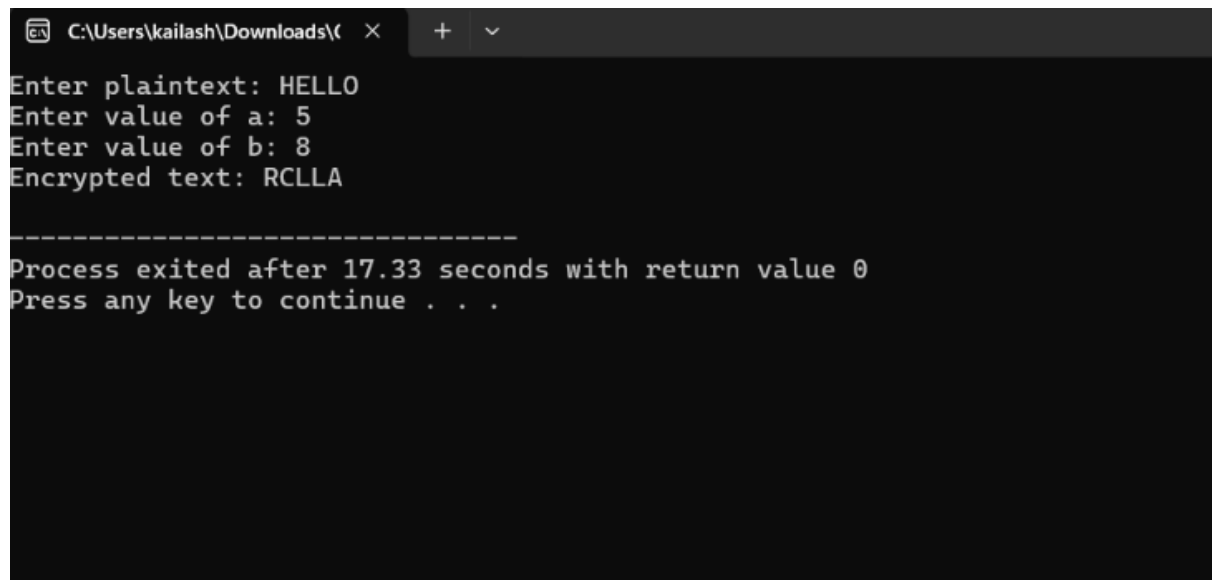
```
        printf("%c", text[i]);
```

```
    }
```

```
}
```

```
printf("\n");  
return 0;  
}
```

Output:



```
C:\Users\kailash\Downloads\> Enter plaintext: HELLO  
Enter value of a: 5  
Enter value of b: 8  
Encrypted text: RCLLA  
-----  
Process exited after 17.33 seconds with return value 0  
Press any key to continue . . .
```